Gold Future/ETF Analysis

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Table 1: A summary of the regression coefficients and measures of goodness of fit for regressing one-day returns of 1, 2, 6 and 12-month futures on 1-day returns of spot gold.

Maturity	**Slope**	**Intercept**	R^2	**RMSE**
1-Month	0.98218	-9.0480×10^{-5}	0.92459	0.00269
2-Month	0.98709	-1.2547×10^{-4}	0.92244	0.00275
6-Month	0.98870	-9.0354×10^{-5}	0.91718	0.00285
12-Month	0.92794	9.3489×10^{-5}	0.76020	0.00520

Table 2: A summary of the slopes and R^2 from the regressions of futures returns versus gold returns over different holding periods.

	Days	**1-Month**	**2-Month**	**6-Month**	**12-Month**
Slope	1	0.96061	0.96608	0.96757	0.91176
	5	0.99723	1.00073	0.99971	0.94958
	10	0.99556	0.99782	0.99821	0.97219
	15	0.99821	1.00003	0.99906	0.98087
R^2	1	0.87699	0.87470	0.87272	0.73603
	5	0.96413	0.96345	0.96008	0.87631
	10	0.98020	0.97931	0.97586	0.93513
	15	0.98603	0.98532	0.98191	0.95316

The out of sample period is defined as starting on January 1, 2024 and is approximately 10% of the data. The RMSEs are calculated in dollar terms assuming a notional investment of $1000 in the spot gold price. The weights are calculated using a constrained least squares regression that forces the weights to sum to one. The GLD RMSE is $6.00 for the out of sample period.

The t-statistics far from zero for UGL and GLL indicate that the slopes are statistically significantly different from the specified leverage ratios.

Table 3: A summary of the weights and in/out of sample RMSEs for portfolios of 1, 2, 3, and 4 futures contracts.

Portfolio	w_0	w_1	w_2	w_6	w_{12}	$\mathbf{RMSE_{in}}$	$\mathbf{RMSE_{out}}$
1-m	0.00285	0.99711				\$3.27	\$9.18
2-m	0.00723		0.99273			\$4.22	\$10.64
12-m	0.04846				0.95208	\$19.36	\$24.98
1-m, 2-m	0.00189	1.22277	-0.22470			\$3.23	\$8.91
1-m, 6-m	0.00198	1.04254		-0.04456		\$3.24	\$8.90
1-m, 12-m	0.00234	1.01049			-0.01286	\$3.26	\$9.16
2-m, 6-m	0.00367		1.22476	-0.22854		\$3.81	\$9.41
2-m, 12-m	0.00390		1.08780		-0.09172	\$3.87	\$10.60
6-m, 12-m	0.00919			1.66574	-0.67435	\$7.00	\$23.51
1-m, 2-m, 6-m	0.00184	1.20060	-0.19291	-0.00956		\$3.23	\$8.88
1-m, 2-m, 12-m	0.00204	1.28345	-0.29678		0.01126	\$3.22	\$8.84
1-m, 6-m, 12-m	0.00210	1.10380		-0.17056	0.06457	\$3.21	\$8.31
2-m, 6-m, 12-m	0.00369		1.22868	-0.23595	0.00348	\$3.81	\$9.37
1-m, 2-m, 6-m, 12-m	0.00196	1.24924	-0.17947	-0.13469	0.06287	\$3.20	\$8.30

Table 4: A summary of the regression coefficients and measures of goodness of fit for regressing one-day returns of (L)ETFs versus spot gold. We include columns for the t-statistic and p-value for testing the hypothesis that the slope equals the specified leverage ratio in each case.

(L)ETF	**Leverage** (β)	**Slope**	**Intercept**	**t-stat**	**p-value**	R^2	**RMSE**
GLD	1	1.00061	-8.8911×10^{-6}	0.25954	0.79523	0.97140	0.00185
UGL	2	2.02088	-1.9483×10^{-4}	2.77734	0.00552	0.96438	0.00349
GLL	-2	-2.01429	1.8890×10^{-4}	-1.92313	0.05457	0.96497	0.00345
DGLD	-2	-1.97565	-8.4896×10^{-5}	0.46841	0.63954	0.40656	0.02098
UGLDF	3	3.00632	4.5874×10^{-4}	0.05822	0.95358	0.25518	0.04463

Table 5: A summary of the weights and in/out of sample RMSEs for portfolios replicating leveraged benchmarks.

Benchmark	**Portfolio**	w_0	w_1	w_2	w_6	w_{12}	$\mathbf{RMSE_{in}}$	$\mathbf{RMSE_{out}}$
2x Leveraged	1-m	-1.62098	2.62436				$58.87	$331.76
	2-m	-1.61086		2.61411			$56.14	$336.05
	6-m	-1.56497			2.56792		$45.85	$357.43
	12-m	-1.49557				2.49767	$46.46	$375.93
	1-m, 2-m	-1.56365	-10.84072	13.40753			$50.70	$353.93
	1-m, 6-m	-1.54357	-1.39403		3.94189		$45.17	$365.64
3x Leveraged	1-m	-4.03105	5.03349				$188.10	$1036.37
	2-m	-4.01308		5.01515			$183.04	$1042.61
	6-m	-3.92928			4.92945		$161.06	$1076.52
	12-m	-3.79391				4.79256	$147.85	$1116.16
	1-m, 2-m	-3.85870	-35.37062	40.23119			$165.26	$1085.79
	1-m, 6-m	-3.79649	-7.19724		11.99866		$149.02	$1111.73